

if
statements

CSCI
134



let's go to
vegas



craps is a popular dice game

the rules of craps

phase one:

- ▶ the player rolls two standard dice
- ▶ if the total is 7 or 11, then the player wins
- ▶ if the total is 2, 3, or 12, then the player loses
- ▶ otherwise, we refer to the total as the "point" and proceed to **phase two**

phase two

- ▶ the player continues to roll until the dice is 7 (in which case, the player loses) or until the dice total equals the point (in which case, the player wins)

the rules of craps



phase one:

- ▶ the player rolls two standard dice
- ▶ if the total is 7 or 11, then the player wins
- ▶ if the total is 2, 3, or 12, then the player loses
- ▶ otherwise, we refer to the total as the "point" and proceed to **phase two**

let's reproduce **phase one**
using python

the player **rolls** two
standard dice

if the **total** is 7 or 11,
then the player wins

if the **total** is 2, 3, or 12,
then the player loses

otherwise, we refer to the
total as the "**point**" and
proceed to phase two

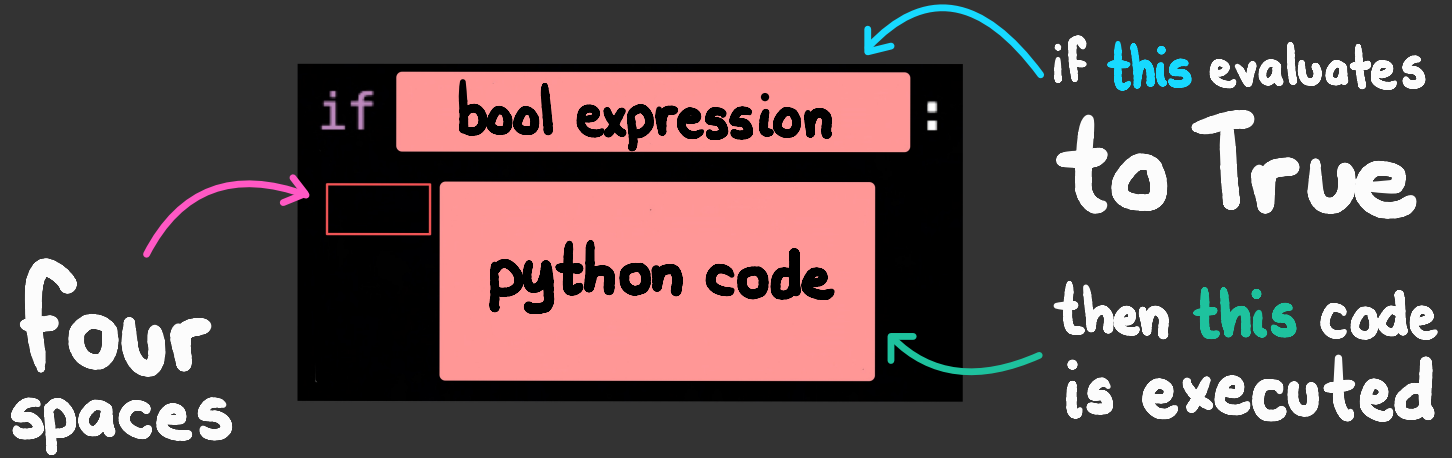
```
die1 = roll()
die2 = roll()
total = die1 + die2
print("You rolled a " + str(total))
if total == 7:
|   print("win!")
elif total == 11:
|   print("win!")
elif total == 2:
|   print("lose!")
elif total == 3:
|   print("lose!")
elif total == 12:
|   print("lose!")
else:
|   point = total
|   print("proceed to phase two")
```

the player **rolls** two
standard dice

```
from random import randint  
  
def roll():  
    return randint(1,6)
```

```
die1 = roll()  
die2 = roll()  
total = die1 + die2  
print("You rolled a " + str(total))  
if total == 7:  
    print("win!")  
elif total == 11:  
    print("win!")  
elif total == 2:  
    print("lose!")  
elif total == 3:  
    print("lose!")  
elif total == 12:  
    print("lose!")  
else:  
    point = total  
    print("proceed to phase two")
```

the **if** statement



the **if** statement
(extended version)

four
spaces

```
if bool expression :  
    python code  
  
elif bool expression :  
    python code  
  
else:  
    python code
```

if **this** evaluates
to **True**

then **this** code
is executed

otherwise,
if **this** evaluates
to **True**

then **this** code
is executed

otherwise,
this code is
executed

the `if` statement (extended version)

```
if bool expression :
```

```
    
```

```
python code
```

```
elif bool expression :
```

```
    
```

```
python code
```

as many `elif`
clauses as
you like
(including zero)

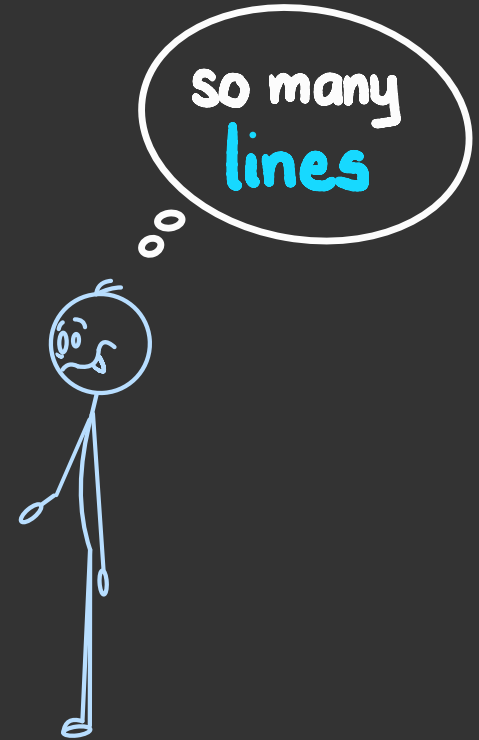
```
else:
```

```
    
```

```
python code
```

at most one
`else` clause per
if statement

```
die1 = roll()
die2 = roll()
total = die1 + die2
print("You rolled a " + str(total))
if total == 7:
|   print("win!")
elif total == 11:
|   print("win!")
elif total == 2:
|   print("lose!")
elif total == 3:
|   print("lose!")
elif total == 12:
|   print("lose!")
else:
|   point = total
|   print("proceed to phase two")
```

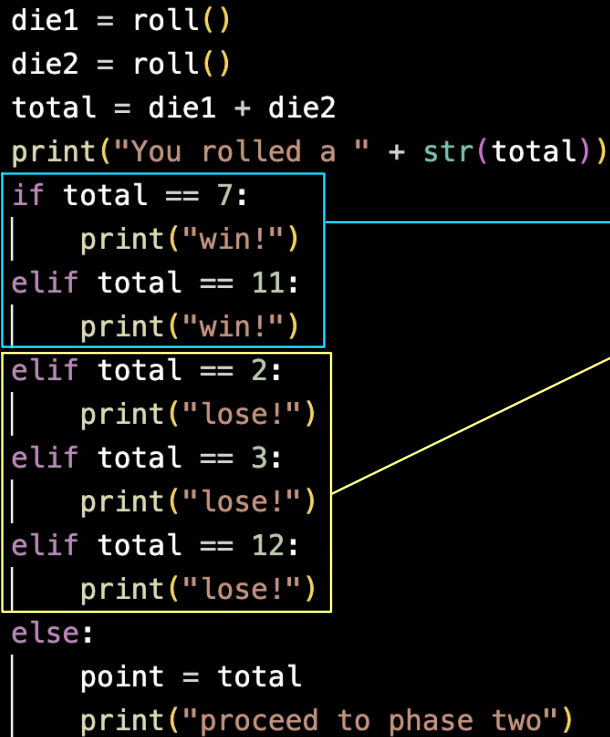


```
die1 = roll()
die2 = roll()
total = die1 + die2
print("You rolled a " + str(total))

if total == 7:
    print("win!")
elif total == 11:
    print("win!")

elif total == 2:
    print("lose!")
elif total == 3:
    print("lose!")
elif total == 12:
    print("lose!")

else:
    point = total
    print("proceed to phase two")
```

A diagram illustrating the use of logical operators in Python. It shows two code snippets. The left snippet is a standard if-elif-else structure. The right snippet uses logical 'or' operators to combine conditions. A blue box highlights the first 'if' condition in the left snippet and the corresponding 'if' condition in the right snippet. A yellow box highlights the subsequent 'elif' conditions in the left snippet and the corresponding 'elif' condition in the right snippet. A yellow line connects the two boxes, indicating the transformation of the logic.

```
die1 = roll()
die2 = roll()
total = die1 + die2
print("You rolled a " + str(total))

if total == 7 or total == 11:
    print("win!")

elif total == 2 or total == 3 or total == 12:
    print("lose!")

else:
    point = total
    print("proceed to phase two")
```

a good use for
logical operators

for what values of **total**
do these programs behave differently?

```
if total <= 3 or total == 12:  
    print("The player loses!")  
elif total <= 6:  
    print("The point has been set!")  
elif total > 7 and total < 11:  
    print("The point has been set!")  
else:  
    print("The player wins!")
```

```
if total <= 3 or total == 12:  
    print("The player loses!")  
if total <= 6:  
    print("The point has been set!")  
if total > 7 and total < 11:  
    print("The point has been set!")  
else:  
    print("The player wins!")
```

total

2,3

4,5,6

12

```
if total <= 3 or total == 12:  
    print("The player loses!")  
elif total <= 6:  
    print("The point has been set!")  
elif total > 7 and total < 11:  
    print("The point has been set!")  
else:  
    print("The player wins!")
```

The player loses!

The point has been set!

The player loses!

```
if total <= 3 or total == 12:  
    print("The player loses!")  
if total <= 6:  
    print("The point has been set!")  
if total > 7 and total < 11:  
    print("The point has been set!")  
else:  
    print("The player wins!")
```

The player loses!
The point has been set!
The player wins!

The point has been set!
The player wins!

The player loses!
The player wins!

statements so far

assignment

variable = expression

def

```
def name (arguments):
```

4 spaces

```
    return return value  
            (what the function computes)
```

for

```
for variable in range(int, int):
```

python code

if

```
if bool expression:
```

python code